

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Specification for Barred Tee
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008
Document Revision	A01
Document Status	Issued For Approval
Originator	Emeka Nwachukwu
Security Classification	Restricted
Issue Date	16-Dec-22

Specification for Barred Tee

A01	16.12.22	Issued for Approval	E. Nwachukwu	M. Ekpunobi	A. Duru	
R01	17.05.2	Issued for Review	E. Uwaoma	E. Nwachukwu	A. Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	17.05.22	Issued for Review
A01	16.12.22	Issued for Approval

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phase, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The document, in consideration of relational design elements & concepts, forms the technical basis for the development of the facilities Project Specifications & tendering of contracts for the project's implementation.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: Specification for Barred Tee

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-001	Specification for Line Pipe
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008	Data Sheets for Barred Tee
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-010	Pipeline Cleaning, Gauging, & Hydrotesting Specification.
EA-BFPCL-GPMC-GGPL-FEED-PPL-PHY-001	Pipeline Integrity Management Philosophy.
	Process Engineering Flow Scheme

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Abbreviations

ABBREVIATIONS	MEANING
ASME	American Society of Mechanical Engineers
API	American Petroleum Institute
CS	Carbon Steel
DEP	Design and Engineering Practices
DP	Design Pressure
FEED	Front End Engineering
HAZ	Heat affected zone
ISO	International Organisation for Standardization
3LPE	Three-Layer Polyethylene
MMscf/d	Million standard cubic feet per day
NDE	Non-Destructive Examination
NPS	Nominal Pipe Size
OD	Outer Diameter
PE	Polyethylene
PSL	Product Specification Level
TD	Design Temperature
SMTS	Specified Minimum Tensile Strength
SMYS	Specified Minimum Yield Strength
WT	Wall thickness
WPS	Welding Procedure Specification
WPQR	Welding Procedure Qualification Record

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

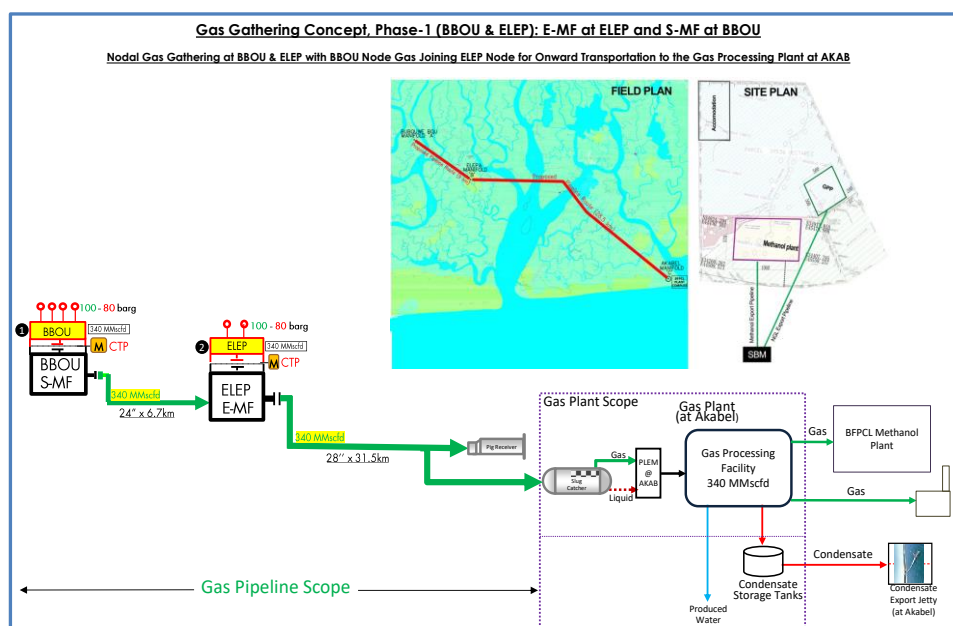


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This specification details the minimum requirements for design, manufacturing, inspection, testing, marking and delivery of carbon steel reducing Barred Tees for phase 1 of BFPCL Gas Gathering Pipelines & Associated Infrastructure.

2. Regulations, Codes And Standards

Regulation	Function
Petroleum Act, CAP P10, Laws of the Federation of Nigeria, LFN 2004, amended.	This is the principal statute regulating the oil industry. The following regulation relevant to this project is issued pursuant to the Act.
Mineral Oil (Safety) Regulations 1963, as amended.	These regulations, in so far as they apply to pipelines, require compliance with relevant IP, API and ASME codes and standards.
Oil Pipelines Act, 1956. CAP. 145 (Amended 1965, CAP. 338, LFN 1990)	Provides for licenses to be granted for the establishment and maintenance of pipelines incidental and supplementary to oil-fields and oil mining, and for purposes ancillary to such pipelines.
Oil and Gas Pipelines Regulations, 1995	Details the regulations for the design, construction and maintenance of oil and gas pipelines.
Environmental Impact Assessment Act, CAP E12, LFN 2004.	Makes the conduct of an EIA mandatory prior to the development of any project or activity likely to have a significant effect on the environment
Federal Environment Protection Agency Act, CAP F13, LFN 2004.	Provides for the protection of the environment and sustainable development of Nigeria's natural resources.
Environmental Guidelines and Standards For The Petroleum Industry In Nigeria, 1991.	DPR (NUPRC) issued document providing guidelines on pollution abatement and relevant legislation.
Local Content Requirements	The provisions of the Nigerian Content Act apply to this project. The design includes strategies to comply with the Act and to maximize Nigerian local content throughout the life of the project. Exemptions / waivers have been sought where required.

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs

- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

2.1 Shell Specifications

The following DEPs are applicable:

REF	DOCUMENT NUMBER	TITLE
[8]	DEP. 31.40.20.37-Gen	Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019)
[9]	DEP. 82.00.10.10-Gen	Project Quality Assurance
[10]	DEP. 31.40.40.38-Gen	Hydrostatic Pressure Testing of new Pipeline
[11]	DEP 31.40.00.10-Gen	Pipeline Engineering (Amendments/Supplements To ISO 13623)
[12]	DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems

2.2 International Codes and Standards

REF	DOCUMENT NUMBER	TITLE
[13]	ASME B31.8	Gas Transmission and Distribution Piping Systems
[14]	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
[15]	API 5L	Specification for Line Pipe
[16]	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
[17]	API RP 5LW	Recommended Practice for Transportation of Line pipe on Barges and Marine Vessels
[18]	API RP 5L1	Recommended Practice for Railroad Transportation of Line Pipe
[19]	BS EN 10204	Metallic Products - Types of Inspection Documents
[20]	ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.
[21]	ISO 9001	Quality Systems Model for Quality Assurance in Design, Development, Production, Installation and Servicing
[22]	API STD 1104	Welding of Pipelines and Related Facilities
[23]	ASME VIII Division 1	Boiler and Pressure Vessel Code – Section VIII: Rules of Construction for Pressure Vessel

[24]	ASME V	Boiler and Pressure Vessel Code – Section V: Non-Destructive Examination
[25]	ASME IX	Boiler and Pressure Vessel Code – Section IX: Welding and Brazing Qualifications
[26]	ASME B16.25	Butt Welding Ends
[27]	ASME B16.9	Factory-Made Wrought Butt-Welding Fittings
[28]	ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
[29]	ASTM A694	Standard Specification for Carbon and Alloy Steel Forgings for Pipe Flanges, Fittings, Valves and Parts for High-Pressure Transmission Service
[30]	ASTM E709	Standard Guide for Magnetic Particle Examination
[31]	BS EN 10204	Metallic Products – Types of Inspection Document
[32]	ISO 9712	Non-Destructive Testing – Qualification and Certification of Personnel
[33]	API Specification 20D	Nondestructive Examination (NDE) Services for Equipment Manufactured in accordance with API Product Specifications used in the Petroleum and Natural Gas Industry

3 Line Pipe Data

3.1 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength (SMYS)	483 Mpa
NPS	24"
Ultimate Tensile Strength	570 MPa
OD / Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

3.2 36" Multiphase Pipeline Design Data

The 28" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.

Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Ultimate Tensile Strength	570MPa
OD / Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Cathodic Protection	Impressed current

4 Design Requirements

4.1 General

Tees shall be designed to meet the requirements of DEP 31.40.00.10-Gen and ASME B31.8. Tees shall be designed to code to withstand the forces and moments equivalent to the base pipe.

All tees shall be barred design, suitable for pigging in either direction through the run of the tee. Reference shall be made to Attachment I, Figure I.2 – Barred Tee Design Requirements.

The bars shall be of the same material type and grade as the tee itself and shall be welded into the branch of the tee via a carrier ring that is seal welded. Bars shall be full penetration welded into the outlet branch after fabrication of the tee.

The design of the bars and their attaching welds shall be such that they can sustain a bi-directional uniformly distributed load equivalent to the full test pressure of the tee multiplied by the internal cross-sectional area of the branch piping. (This load is equivalent to that which would be seen if a pig were to become trapped in the branch of the tee with full test pressure on one side and no pressure on the other).

Where required, the bars shall be recessed into the branch of the tee to sufficient depth to prevent their hindering the passage of a pig but at the same time, at not such a depth as to allow a pig to rotate and become trapped in the tee.

4.2 Dimensions

Dimensions and tolerances shall be in accordance with ASME B16.9, unless modified by the data sheet. The diameters and wall thicknesses of the tee at each weld end shall be as shown on the data sheet.

The tolerance on the nominal internal diameter, defined as the nominal external diameter minus twice the nominal wall thickness, shall meet the requirements of ASME B16.9. Out of roundness shall be assessed using an internal ring gauge of diameter 5mm less than the nominal internal diameter and of length equal to 1.5 times its diameter shall be assumed.

A gauge shall pass freely through the barred tee when held normal to the pipe axis.

Weld ends shall be bevelled in accordance with ASME B16.9.

At the branch outlets the maximum wall thickness shall not exceed the nominal wall thickness by more than 10%. Excess material shall be machined back at an angle not exceeding 14° (1 in 4 taper), all in accordance with ASME B31.8 Section 422.1 (a)-(b).

Run and branch connections shall be bevelled in accordance with ASME B16.9, to match the size and wall thickness of the connecting pipes. The taper to accommodate the change in thickness from line pipe shall be on the outside diameter of the fitting only. Internal bores shall match the pipe.

4.3 Design Calculations

The Supplier shall design each tee for the worst combination of temperatures, pressures, forces, and moments. COMPANY is to advise maximum forces and moments on tee.

The calculated wall thickness shall include for a corrosion allowance as specified in Data Sheet for Bar Tees.

Apart from the reinforcement areas, the reducing barred tee ends will have the same nominal dimensions as the pipeline to which it is attached. The matching wall thickness of the line pipe shall be as specified in the data sheet. The mating material shall be in accordance with API 5L X70.

Rules for the design and reinforcement of branch outlets are specified in DEP 31.40.00.10-Gen, and ASME B31.8.

Any computer programs used for the work shall be fully verified.

Supplier shall submit detail calculations for thickness and reinforcement together with drawings to Company for approval prior to start of manufacturing.

5 Material Requirements

5.1 Manufacture

Forged tees shall be manufactured by a hot forging process in accordance with ASTM A694, and shall be suitable for welding to line pipe manufactured in accordance with API 5L X70. Tees shall be forged into shape such that finishing of the forgings may be achieved with the minimum of machining.

Unless otherwise agreed by the COMPANY, all tees shall be from the same heat (cast) of steel and the same heat treatment batch.

Materials used in the manufacture of the tees may be forged or seamless carbon steel pipe material.

Steels shall be made from a basic oxygen process or electrical arc furnace method. Only fully killed fine grain steels shall be used. The steel shall be of carbon or carbon manganese-based composition.

The mechanical testing schedule and specimen location shall be as agreed by the COMPANY. The specimen shall be transverse and taken from mid thickness location for extruded and forged Tee

The mechanical properties of the material shall be as follows:

Yield Strength	483 N/mm ² (minimum)
Tensile Strength	570 N/mm ² (minimum)
Elongation % of 50mm specimen	20% (minimum)

The testing procedure shall be in accordance with ASTM A370,

The impact toughness of the tees shall be determined by Charpy V-notch (CVN) impact testing in accordance with ASTM A370, using full-size specimens where possible. Three Charpy specimens shall be taken from the parent metal at locations agreed by the COMPANY.

The impact test temperature shall be lower than or equal to the minimum design temperature, as specified in the data sheet.

For approval testing, specimen location and orientation shall be longitudinal.

For production quality control testing, specimens may be taken from forging prolongations. The specimen orientation shall be tangential.

The Charpy energy acceptance criterion shown is for full size 10mm x 10mm specimens. Where sub-size specimens are used, the energy acceptance criteria may be adjusted proportionally.

5.2 Heat Treatment

Tees shall be supplied in one of the heat-treated conditions:

- Normalised
- Normalised and Tempered
- Normalised, Quenched and Tempered

Supplier shall advise COMPANY of the heat treatment to be applied with his bid.

Heat treatment shall be such to ensure the mechanical properties specified in Section 6.1. All thermo / mechanical treatment of material shall be subject to review and acceptance by COMPANY.

There shall be at least one thermocouple connected directly to the barred tee during heat treatment.

6 Welding

6.1 General

Welds, including welding of guide bars in tees, shall be made according to procedures qualified to the requirements of ASME IX.

All welds on barred tees for guide bars and attaching guide bars to the branch of the tee shall be full penetration and continuous around the guide bars.

6.2 Defect Repair

No repair work shall proceed without prior written authorisation from the COMPANY. Repairs of forgings by welding shall not be permitted.

Surface defects such as slivers, sharp notches, gouges, scores, pits and severe machining marks shall be considered injurious defects and in need of repair. Minor surface defects may be removed by machining or grinding, where the wall thickness is not reduced locally by more than 5% of the nominal thickness, and when the minimum specified wall thickness is maintained. The remaining wall thickness shall be verified to the satisfaction of the COMPANY using ultrasonic or other calibrated instruments.

The machined or ground area(s) shall be carefully dressed to ensure smooth transition with the surrounding surface of the forging with no sharp notches or contours.

The machined or ground area(s) shall be examined for any defects using magnetic particle examination. Magnetic Particle Inspection (MPI) shall be carried out in accordance with ASME V Article 7 / ASTM E709.

If the section thickness is reduced below the minimum section thickness required in accordance with this Specification, the tee shall be rejected.

Cracks shall not be repaired and shall be cause for rejection. The cause of cracking shall be investigated and established to the satisfaction of the COMPANY.

7 Testing And Inspection

7.1 General

Supplier will carry out inspection as detailed in this Specification. This inspection does not remove Supplier from the responsibility to perform inspection activities sufficient to ensure the quality of the product.

Supplier inspectors and equipment are subject to the approval of COMPANY. COMPANY will be the sole judge of the acceptability of materials and/or work done.

Personnel carrying out non-destructive inspection shall be experienced in using the equipment specified and shall be qualified as per API Specification 20D and ASNT SNT-TC-1A Level II, ISO 9712.

The person responsible for the preparation of inspection procedures and supervision of inspection activities shall be qualified as per API Specification 20D and ASNT SNT-TC-1A Level II, ISO 9712.

7.2 Mechanical Testing

Mechanical tests for the tees shall be carried out on test pieces cut from full size test forging in accordance with COMPANY accepted locations shown in Attachment I.

Mechanical tests shall be carried out on test pieces cut from a test item of each material grade, heat treatment batch. For small items from which a complete suite of test pieces cannot be cut, the Supplier shall propose test locations that can provide the requisite mechanical test data as appropriate to the specific geometry.

7.3 Re-Tests

If any of the results does not conform to the specified requirements, the Supplier may heat-treat the batch forgings again provided they have COMPANY acceptance. The material shall then be tested in accordance with this Specification.

7.4 Visual Inspection

All tees, including bevelled ends, shall be subject to 100% visual inspection to ensure freedom from surface defects and compliance with dimensional tolerances.

7.5 Magnetic Particle Inspection

100% magnetic particle examination shall be carried out on all external surfaces of the finished tee after heat treatment and final machining.

Magnetic Particle Inspection (MPI) shall be carried out as follows:

Procedure: API Specification 20D and ASME V Article 7/ ASTM E709.

Acceptance: API Specification 20D and ASME VIII Div. 1 Appendix 6.

No indications will be permitted on bevelled ends. All indications shall be removed by cutting back or grinding and subject to re-examination by magnetic particle examination.

Acceptance criteria for girth welds, such as those for attachment of pup-piece to fittings shall be in accordance with API STD 1104.

7.6 Ultrasonic Examination

The tees shall be subject to 100% ultrasonic examination which shall be carried out as follows:

Procedure: API Specification 20D and ASME V Article 5.

Acceptance: API Specification 20D and ASME VIII Div. 1 Appendix 12.

7.7 Hydrostatic testing

The Supplier shall certify to the COMPANY that each tee is capable of withstanding, without leaks, ruptures or unacceptable deformation, the pipeline hydrostatic test pressure as specified on the Data Sheets for Barred Tees.

8 Marking

8.1 Identification

The tees shall be marked in accordance with ANSI B16.9. Permanent marking shall be by low stress or dot dye hard stamping or vibro etching.

Each tee shall be identified by markings, as a minimum of 25mm in height, applied in the sequence detailed below:

- COMPANY and Project Name
- Suppliers Name
- Purchase Order No. and Purchase Order Item No.
- Tag / SP Number
- Material Grade
- Design Pressure
- Nominal diameter and wall thickness
- Heat number

Any fitting Quenched and Tempered during manufacture shall be marked 'QT' and HARD STAMPED using round nose low stress stamping. Details of temperatures and times of heat treatments shall be recorded on the certification.

9 Handling, Packing And Transportation

Manufacturer shall submit to COMPANY for approval, details of proposed methods of handling, storage, packing and shipping of the Reducing Barred Tees.

End caps and / or bevel protectors shall be fitted to the Reducing Barred Tees after completion of individual manufacturing processes.

All handling shall be done such that mechanical damage to the Reducing Barred Tees is avoided during loading / offloading or while in transit.

The Reducing Barred Tees shall be treated with a colourless resin to protect against surface rust during transit and storage before installation. The resin used shall be such as to be easily removed by regular solvents prior to installation.

Barred Tees are to be arranged such as to avoid water retention. Storage shall take place in a specially designated area, where the material will be stacked in a safe, orderly manner, to eliminate the risk of damage. Protection and packing of Reducing Barred Tees shall be performed before shipment. All itemized parts of the order shall be numbered, countersigned, and marked accordingly.

Rail cars, trucks, lighters, ships, or any other conveyance used for transportation of the Reducing Barred Tees shall be cleaned of debris, or any other item or substance, that may possibly contaminate or damage them during loading / off-loading or whilst in transit.

10 Quality Assurance And Quality Control

Supplier shall show that an effective Quality Management System is in operation for both products and services and this shall comply with ISO 9000 family Standards.

The Supplier shall establish and maintain a fully documented Quality Assurance / Quality Control system in accordance with ISO 9000 family Standards to ensure:

- Effective inspection and objective evidence that items conform to Purchase Order requirements
- Adequate identification and suitable handling of items
- Determination of customer needs
- Definition of the method to be employed to achieve them
- Assessment of the criticality of activities and products
- Allocation of responsibility and definition of controls
- Generation and maintenance of quality records
- Quality improvement.

The Supplier shall submit documentary evidence of any Quality Reviews to ISO 9000 family Standards and internal system audits planning and records and NCR / CAR.

All certification, test results, reports and any other documentation submitted to the COMPANY relative to the inspection system and results of inspections shall be in the English language.

Any verification of quality performed by COMPANY or its representative will not relieve the Supplier in any way of its responsibility to produce acceptable items nor shall it preclude any subsequent rejection of items.

COMPANY shall establish a Quality Plan and Inspection Test Plan with inspection points and mandatory hold points where applicable, which require verification of selected characteristics of any items or process by COMPANY or agreed Third Party and beyond which the manufacture shall not proceed until released.

The Supplier shall establish and maintain a system of traceability, which complies with ISO 9000 family Standards. The System shall be such that each line pipe is allocated a unique number, and this number shall be maintained such that at any point the individual line pipe can be traced back to its original heat number. These numbers will be in blocks allocated by the COMPANY to the

Supplier, and the Supplier shall demonstrate its ability to maintain traceability of such numbers throughout production.

11 Documentation

Warranty provisions shall be specified in the Standard Terms and Conditions attached to the Purchase Order and shall apply to the Supplier, as well as the Supplier's subcontractors. Where modified warranty terms are agreed to between COMPANY and the Supplier, these modified provisions will be incorporated in the Purchase Order.

All certification and documentation shall be as per ISO 3183-3 certification requirements.

Barred Tees shall be delivered by the Supplier with certificates from a classification society in accordance with BS EN 10204 3.2. The certificates and associated documentation supplied to COMPANY shall contain all relevant information, including, but not restricted to the following:

- Supplier's Identification
- COMPANY's Name and Order Number
- Metallographic Examination Records
- Description of Product, Dimension, Identification Number, Weight
- Steel Making Process
- Chemical Composition
- Heat Number
- Identification of Steel type and Grade
- Yield - Tensile Ratio
- Charpy Impact Results
- Hardness Tests
- Heat Treatment Condition and Records
- Hydrostatic Test
- NDE Procedures and Results
- Surface Inspection
- Dimensional Control Checks
- Manufacturing Procedure and Qualification Tests
- Supplier's Third Party BS EN 10204, 3.2 Certificates

The Supplier shall also supply a table of information stating the heat number, length, weight, testing, and inspection associated with each barred tee number.

12 Attachments

12.1 Attachment I – Tee Design and Testing Requirements

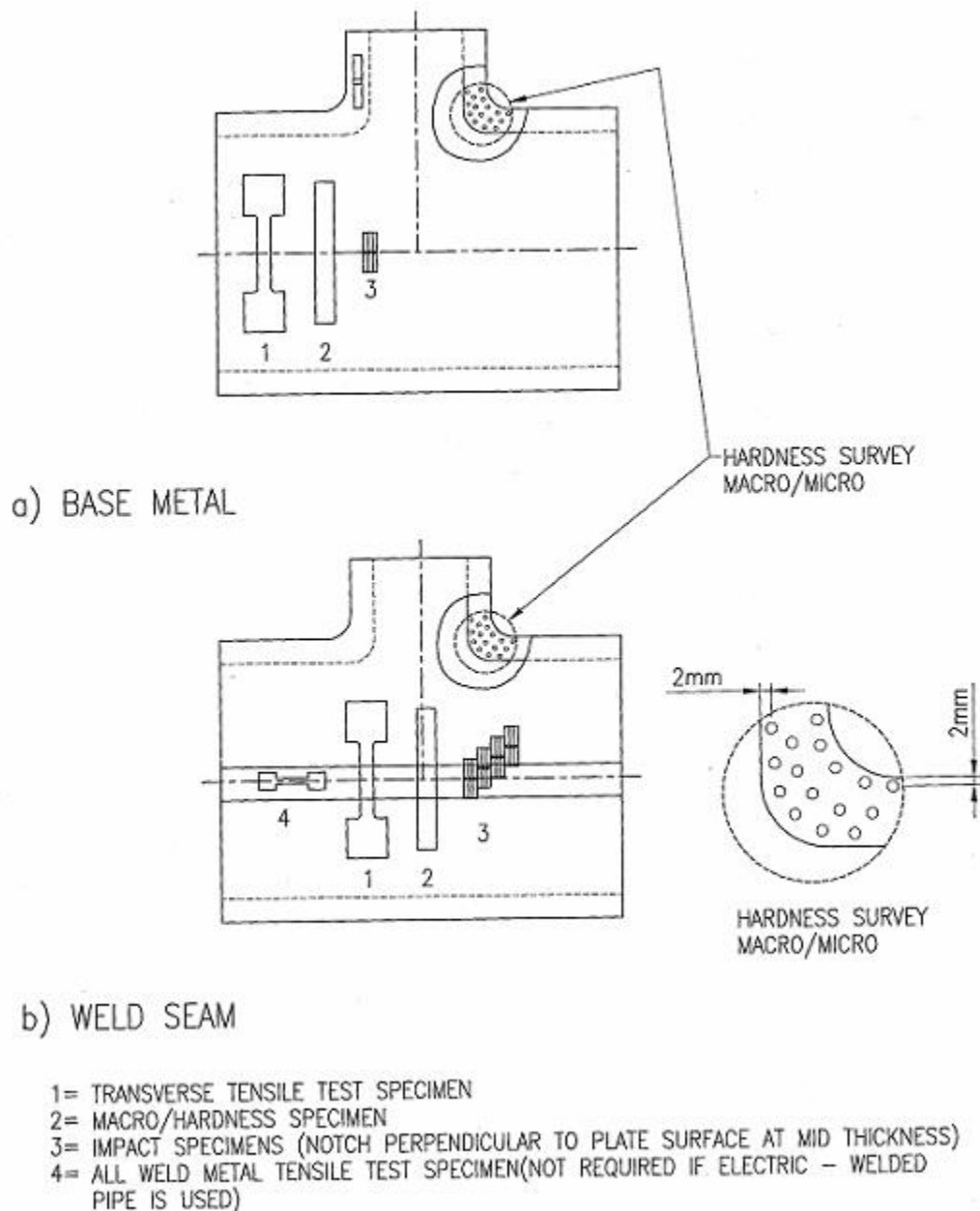


Figure I.1: Mechanical Test Location

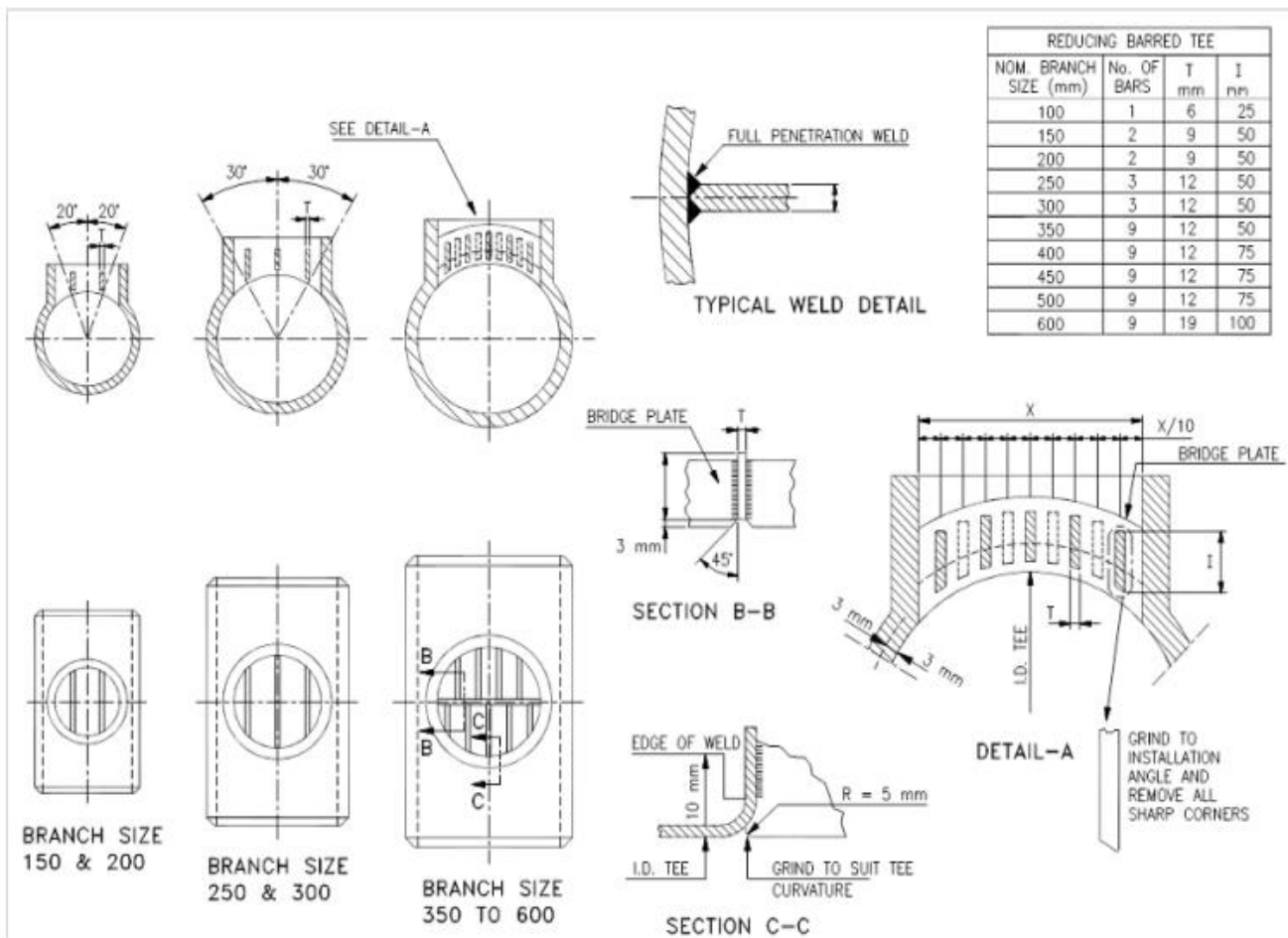


Figure I.2: Barred Tee Design Requirement